Chapter 20 Maxwell's Equations

20.1 Quasi-equations and monopole

$$\oint \boldsymbol{D} \cdot d\boldsymbol{S} = q$$

$$\oint \boldsymbol{B} \cdot d\boldsymbol{S} = \mu_0 g$$

$$\oint \boldsymbol{E} \cdot d\boldsymbol{l} = -\frac{d}{dt}\int \boldsymbol{B} \cdot d\boldsymbol{S} + \mu_0 I_\mathrm{m}$$

$$\oint \boldsymbol{H} \cdot d\boldsymbol{l} = I + \frac{d}{dt}\int \boldsymbol{D} \cdot d\boldsymbol{S}$$

If monopole g is a reality, we have

$$\oint \boldsymbol{B} \cdot d\boldsymbol{S} = \mu_0 g$$

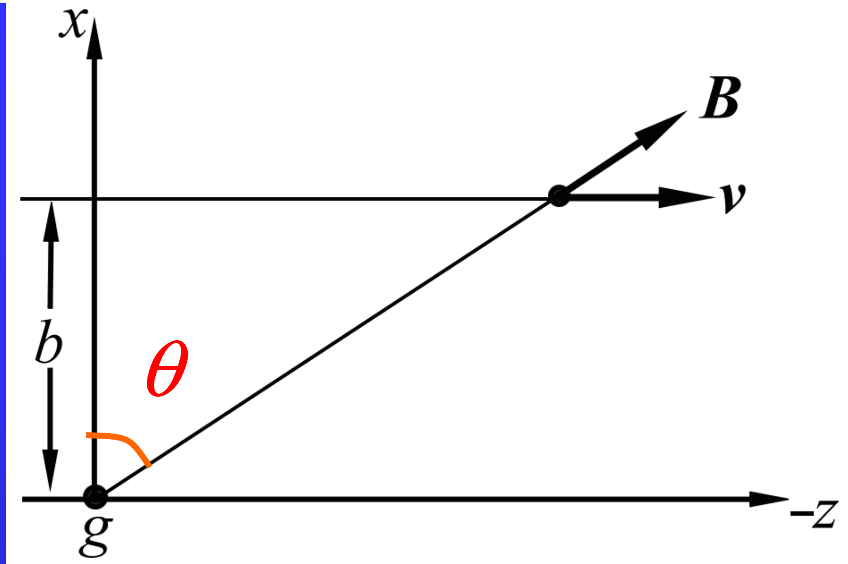
Then ,the magnetic induction is

$$\boldsymbol{B} = \frac{\mu_0 g}{4\pi} \frac{\boldsymbol{r}}{r^3}.$$

$$\sim \boldsymbol{E} = \frac{1}{4\pi\varepsilon_0} \frac{e}{r^3} \boldsymbol{r}$$

$$F_y = evB_x$$

$$= ev\frac{\mu_0}{4\pi}\frac{g}{(b^2+v^2t^2)}\frac{b}{(b^2+v^2t^2)^{1/2}}.$$



$$\Delta p_y = \int F_y\,dt$$

$$= ev\frac{\mu_0}{4\pi}gb\int_{-\infty}^{\infty}\frac{dt}{(b^2+v^2t^2)^{3/2}}.$$

Set $vt / b = \tan\theta$

$$\Delta p_y = \frac{\mu_0}{4\pi}\frac{eg}{b}\int_{-\pi/2}^{\pi/2}\frac{\sec^2\theta\,d\theta}{\sec^3\theta}$$

$$= \frac{\mu_0}{4\pi}\frac{2eg}{b}.$$

$$\Delta L_z = x\,\Delta p_y = 2e\times\frac{\mu_0}{4\pi}g \quad = n\hbar$$

‘Dirac quantization condition’ proposed by Goldhaber

$$\oint \boldsymbol{B}\cdot d\boldsymbol{S} = \mu_0 g = \frac{4\pi}{2e}2e\times\frac{\mu_0}{4\pi}g = \frac{4\pi n\hbar}{2e} = n\left(2\Phi_0\right).$$

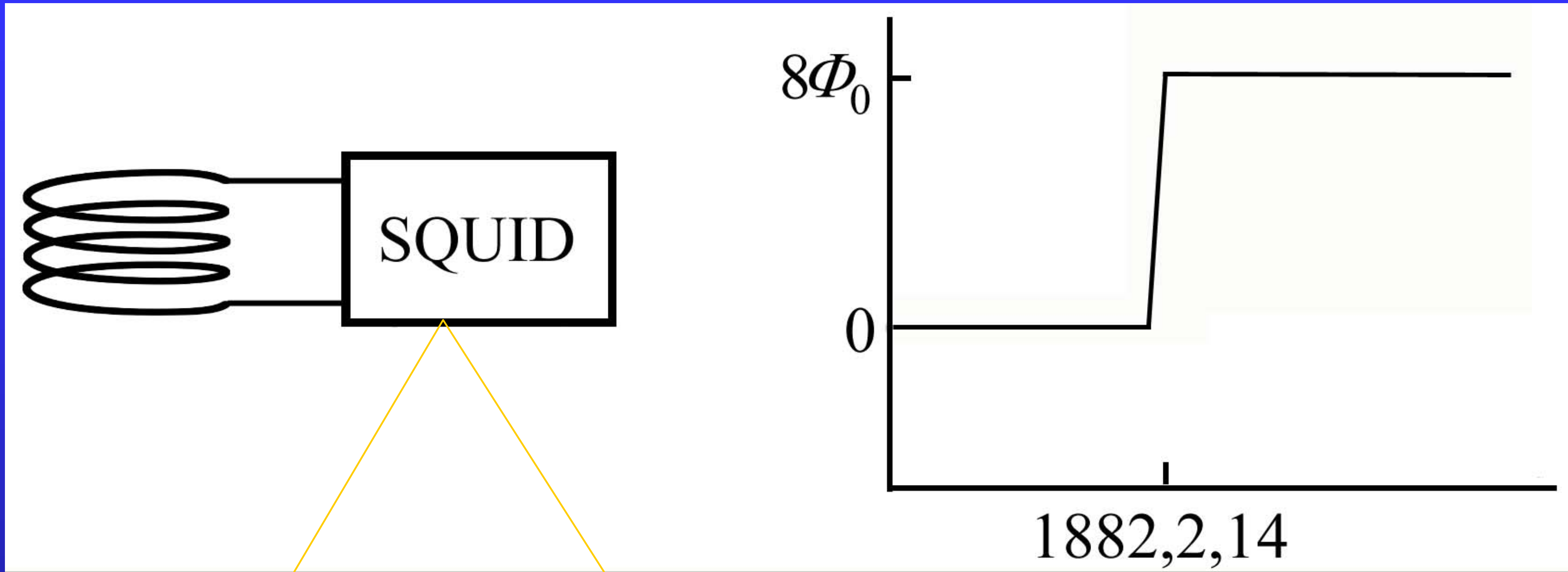
fluxon, flux quantum of superconductivity

$$\Phi_0 = \frac{2\pi\hbar}{2e} = \frac{h}{2e} = 2.07\times10^{-15}\ \mathrm{Wb}$$

$$eg = n\left(\frac{2\pi\hbar}{\mu_0}\right)$$

In 1931 the English physicist P.A.M. Dirac proposed that the existence of even a single magnetic monopole in the universe would explain why electric charge comes only in multiples of the electron charge.

1982 Cabrera et al



superconducting quantum interference device

$$6.1\times10^{-10}\ \mathrm{cm}^{-1}\cdot\mathrm{s}^{-1}\cdot\mathrm{sr}^{-1}$$

steradian

$$3.7\times10^{-11}\ \mathrm{cm}^{-1}\cdot\mathrm{s}^{-1}\cdot\mathrm{sr}^{-1}$$

***The grand unified monopole** (GUM)

magnetic monopoles are a more or less inevitable consequence of GUTs

as heavy as amoeba, onion structure,

To the theorists acute embarrassment there seemed to be a superabundance of GUMs

Guth of MIT invented inflationary scenario

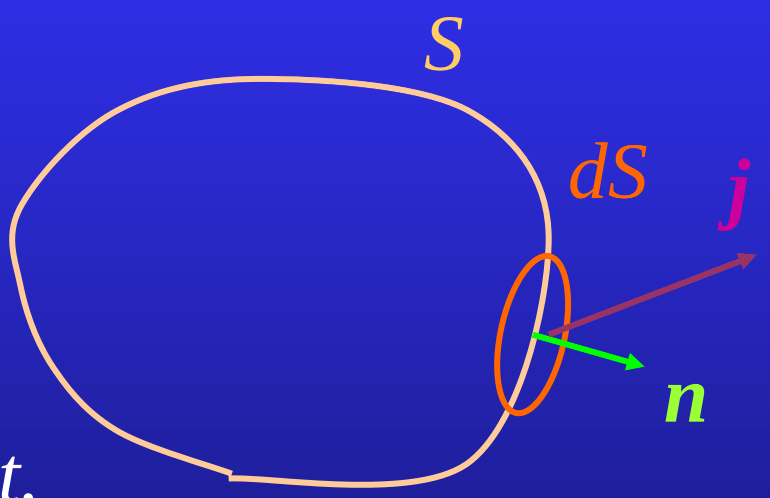
‘magnetic monopole’ 1750 matches on yahoo

20.2 Displacement current and induced magnetic field

The principle of conservation of charge

The current through a closed surface S is

$$I = \oint_S \boldsymbol{j} \cdot d\boldsymbol{S}.$$



Suppose the loss of charge within S per unit time is dq/dt. Thus

$$-\frac{dq}{dt} = \oint_S \boldsymbol{j} \cdot d\boldsymbol{S}.$$

$$q = \int_V \rho\, d\tau, \qquad \oint_S \boldsymbol{j} \cdot d\boldsymbol{S} = \int_V \nabla \cdot \boldsymbol{j}\, d\tau,$$

$$-\frac{dq}{dt} = \oint_S \boldsymbol{j} \cdot d\boldsymbol{S} \Rightarrow \int_V \frac{\partial \rho}{\partial t} d\tau + \int_V \nabla \cdot \boldsymbol{j}\, d\tau = 0.$$

$$\Downarrow$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \boldsymbol{j} = 0.$$

the continuity equation

According to Gauss' law

$$q = \oint_s \boldsymbol{D} \cdot d\boldsymbol{S}.$$

Then

$$\frac{dq}{dt} = \oint_s \frac{\partial \boldsymbol{D}}{\partial t} \cdot d\boldsymbol{S}.$$

Using $-\dfrac{dq}{dt} = \oint_S \boldsymbol{j} \cdot d\boldsymbol{S},$ we have

$$\oint_s (\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}) \cdot d\boldsymbol{S} = 0.$$

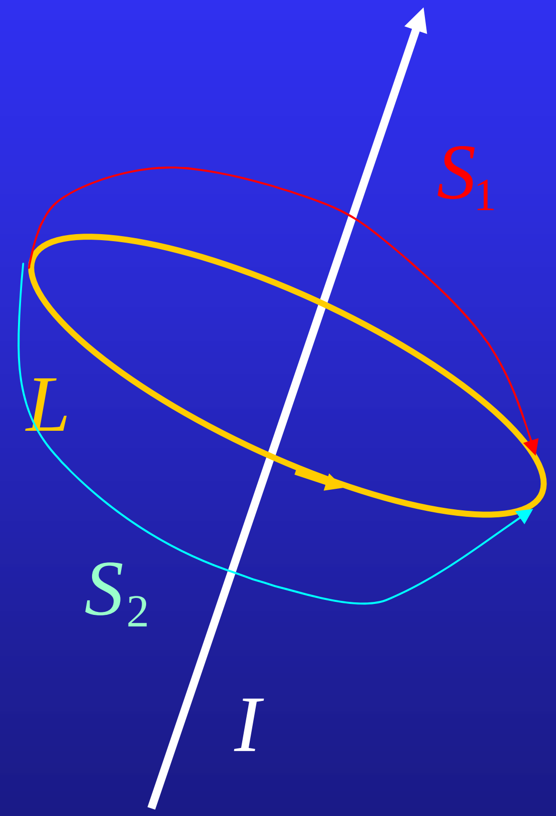
In the steady case, $\dfrac{\partial \boldsymbol{D}}{\partial t} = 0$, then

$$\oint_S \boldsymbol{j} \cdot d\boldsymbol{S} = 0.$$

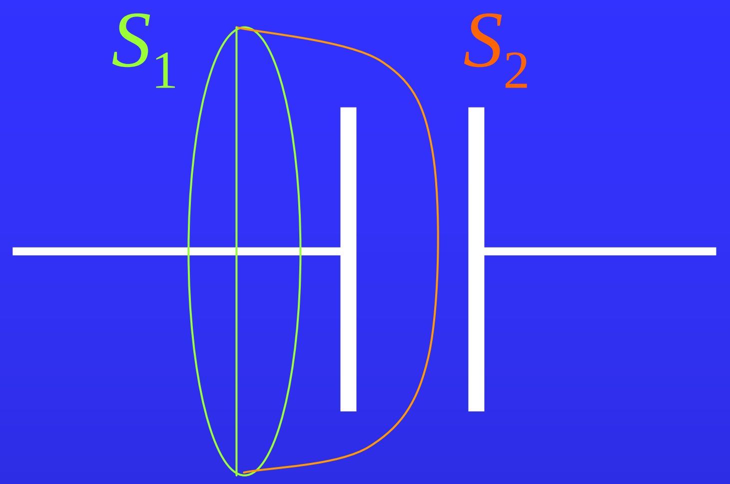
$$\Downarrow$$

$$\oint_L \boldsymbol{H} \cdot d\boldsymbol{l} = \int_S \boldsymbol{j} \cdot d\boldsymbol{S}.$$

$$\int_{S_1} \boldsymbol{j} \cdot d\boldsymbol{S} = \int_{S_2} \boldsymbol{j} \cdot d\boldsymbol{S}$$



In the non-steady case, e.g. the charging and discharging process of a capacitor, the Ampere's law



$$\oint_L \boldsymbol{H} \cdot d\boldsymbol{l} = \int_S \boldsymbol{j} \cdot d\boldsymbol{S}$$

has different results for $S=S_1$ and $S=S_2$. Thus, Ampere's law need to be revised in the non-steady situation.

steady $\oint_S \boldsymbol{j} \cdot d\boldsymbol{S} = 0.$

$$\Downarrow$$

$$\int_{S_1} \boldsymbol{j} \cdot d\boldsymbol{S} = \int_{S_2} \boldsymbol{j} \cdot d\boldsymbol{S}$$

non-steady $\oint_s (\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}) \cdot d\boldsymbol{S} = 0.$

$$\Downarrow$$

$$\int_{S_1} (\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}) \cdot d\boldsymbol{S} = \int_{S_2} (\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}) \cdot d\boldsymbol{S}$$

$$\boldsymbol{j} \to \boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}$$

Ampere-Maxwell law

$$\boxed{\oint_L \boldsymbol{H} \cdot d\boldsymbol{l} = \int_S (\boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}) \cdot d\boldsymbol{S}}$$

$$= \int_S \boldsymbol{j} \cdot d\boldsymbol{S} + \underline{\int_S \frac{\partial \boldsymbol{D}}{\partial t} \cdot d\boldsymbol{S}.}$$

the displacement current

Using Stokes' theorem

$$\oint_L \boldsymbol{H} \cdot d\boldsymbol{l} = \int_S \nabla \times \boldsymbol{H} \cdot d\boldsymbol{S},$$

we obtain

$$\boxed{\nabla \times \boldsymbol{H} = \boldsymbol{j} + \frac{\partial \boldsymbol{D}}{\partial t}.}$$

A-M law
in differential form

In free space $\boldsymbol{j}= 0$, $\boldsymbol{H} = \boldsymbol{B} / \mu_0$, $\boldsymbol{D} = \varepsilon_0 \boldsymbol{E}$, we have

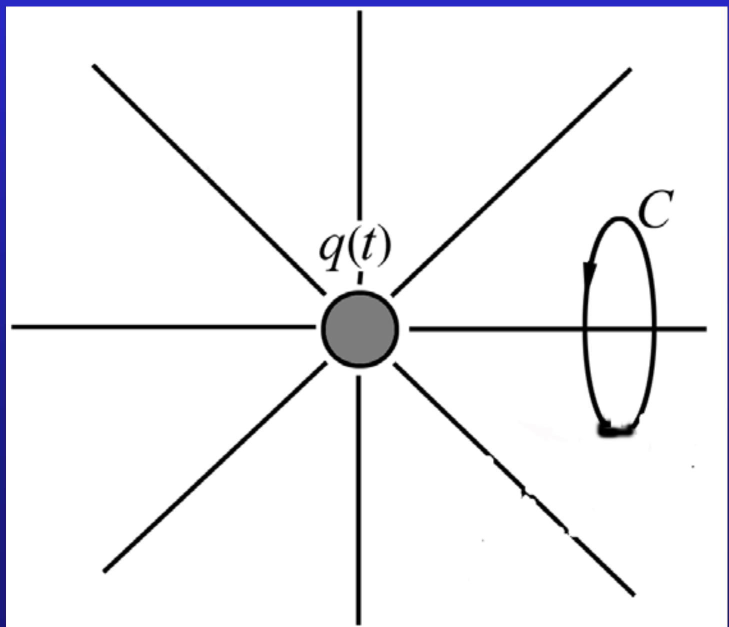
$$\oint_L \boldsymbol{B} \cdot d\boldsymbol{l} = \mu_0 \varepsilon_0 \int_S \boldsymbol{E} \cdot d\boldsymbol{S} = \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{\Phi}_E}{\partial t}$$

Varying electric field induces magnetic field.

Example 20.1 Find the magnetic field of a spherically symmetric charge distribution and an isotopic current.

Solution: The electric current through an arbitrary loop C as shown in the figure is

$$I_C = \int_{S_C} \boldsymbol{j}(\boldsymbol{r}) \cdot d\boldsymbol{S} = \frac{\Omega}{4\pi} \oint \boldsymbol{j}(\boldsymbol{r}) \cdot d\boldsymbol{S} = \frac{\Omega}{4\pi} I,$$



where S_C is a surface that its edge is C , Ω is the solid angle subtended by S_C as viewed from the center of the charge distribution, I is the total current through a closed surface around the charge distribution.

The electric displacement flux through S_C is

$$\int_{S_C} \boldsymbol{D} \cdot d\boldsymbol{S} = \frac{\Omega}{4\pi} \oint \boldsymbol{D} \cdot d\boldsymbol{S}.$$

Then, using Ampere-Maxwell law, we have

$$\oint_C \boldsymbol{H} \cdot d\boldsymbol{l} = I_C + \frac{\partial}{\partial t} \int_{S_C} \boldsymbol{D} \cdot d\boldsymbol{S}$$

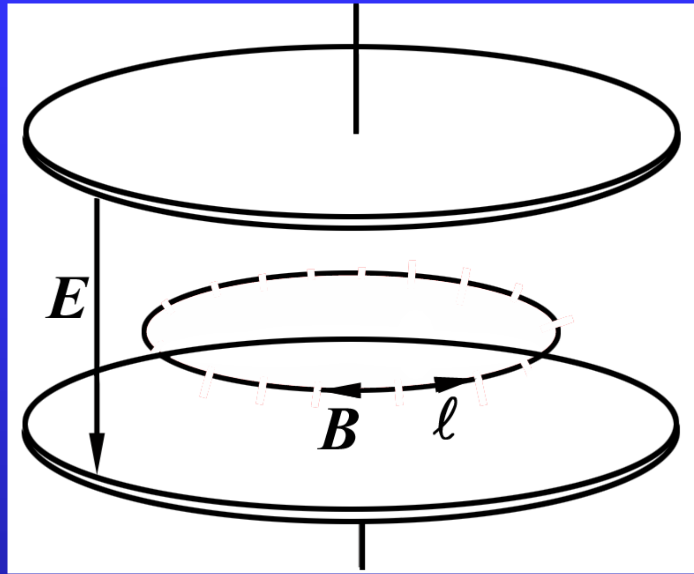
$$= \frac{\Omega}{4\pi} \oint \left(\boldsymbol{j}(\boldsymbol{r}) + \frac{\partial \boldsymbol{D}}{\partial t} \right) \cdot d\boldsymbol{S}$$

$$= \frac{\Omega}{4\pi} \left(I + \frac{\partial q}{\partial t} \right)$$

$$= 0.$$

This implies the magnetic field is zero.

Example 20.2 Find the magnetic field at the edge of a circular parallel-plate capacitor.



Solution: The circulation of the magnetic field around a circle of radius r concentric with the the plate is

$$\oint \boldsymbol{B}\cdot d\boldsymbol{l} = B_\varphi 2\pi r.$$

For $r<R$, we have

$$\mu_0\varepsilon_0 \frac{\partial \Phi_{\mathrm{E}}}{\partial t} = -\mu_0\varepsilon_0 \pi r^2 \frac{\partial E}{\partial t}.$$

For $r\geq R$, we have

$$\mu_0\varepsilon_0 \frac{\partial \Phi_{\mathrm{E}}}{\partial t} = -\varepsilon_0\mu_0\ R^2 \frac{\partial E}{\partial t}.$$

Using Ampere-Maxwell law, the magnetic field is

$$B = \begin{cases} \dfrac{1}{2}\mu_0\varepsilon_0\, r\,\dfrac{\partial E}{\partial t}, & r < R, \\ \dfrac{1}{2r}\mu_0\varepsilon_0\, R^2\,\dfrac{\partial E}{\partial t}, & r \geq R. \end{cases}$$

At the edge of the capacitor, if

$$r = R = 5.0\text{cm}\,,\ \frac{\partial E}{\partial t} = 10^{12}\ \text{V}/(\text{m}\cdot\text{s}),$$

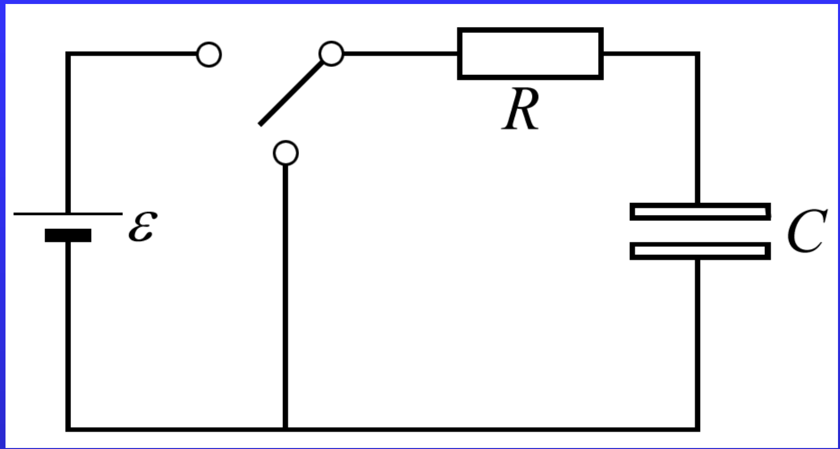
the magnetic field is

$$B = \frac{1}{2}\frac{\mu_0}{4\pi}4\pi\varepsilon_0\, R\frac{\partial E}{\partial t}$$

$$= 0.5\times(3.0\times10^{8})^{-2}\times5.0\times10^{-2}\times10^{12}\ \text{T}$$

$$= 2.8\times10^{-7}\ \text{T} \quad \ll 10^{-4}\text{T}$$

Example 20.3 Displacement current in a *RC* circuit



For the charging and discharging process,

$$\varepsilon = R\frac{dq}{dt} + \frac{q}{C}, q = C\varepsilon\left(1 - e^{-\frac{t}{RC}}\right), I = \frac{\varepsilon}{R}e^{-\frac{t}{RC}}$$

$$0 = R\frac{dq}{dt} + \frac{q}{C}, q = q_0 e^{-\frac{t}{RC}}, I = -\frac{q_0}{RC}e^{-\frac{t}{RC}}$$

In the gap

$$E_{\text{gap}} = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A}$$

$$I_{\text{d}} = \varepsilon_0 \frac{\partial \Phi_{\text{E}}}{\partial t} = \varepsilon_0 \frac{\partial}{\partial t}(AE_{\text{gap}}) = \frac{dq}{dt} = I$$

$$= \frac{4\pi\varepsilon_0}{4} R^2 \frac{\partial E}{\partial t} = 0.070\text{A}$$

$$B_R = \frac{\mu_0}{4\pi}\frac{2I_{\text{d}}}{R} << \frac{\mu_0}{4\pi}\frac{2I}{r_{\text{wire}}} = B_r$$

In the wire, Ohm's law is valid.

$$I_{\mathrm{d}} = \varepsilon_0 \varepsilon_{\mathrm{r}} \frac{\mathrm{d}}{\mathrm{d}t}(EA) = \frac{\varepsilon_0 \varepsilon_{\mathrm{r}}}{\sigma_{\mathrm{c}}} \frac{\mathrm{d}}{\mathrm{d}t} I = \frac{\varepsilon_0 \varepsilon_{\mathrm{r}}}{\sigma_{\mathrm{c}}} \left(-\frac{1}{RC} \right) I$$

$$\frac{\varepsilon_0 \varepsilon_{\mathrm{r}}}{\sigma_{\mathrm{c}}} \sim \frac{\varepsilon_{\mathrm{r}}}{4\pi \times 9 \times 10^9 \times 5.8 \times 10^7} \sim 10^{-18}\,\mathrm{s}$$

$$RC \sim \Omega \cdot \mu\mathrm{F}(\Omega \cdot \mathrm{pF}) \sim 10^{-6}\,\mathrm{s}(10^{-12}\,\mathrm{s})$$

$$\frac{I_{\mathrm{d}}}{I} \sim 10^{-6}$$

Displacement current

Maxwell's equations

Electromagnetic waves

20.3 Maxwell's equations

The entire theory of the electromagnetic field can be condensed into following four laws:

	integral form	differential form
Gauss' law for electric field:	$\int \nabla \cdot \boldsymbol{D}\, d\tau = q_{\text{free}}$,	$\nabla \cdot \boldsymbol{D} = \rho_{\text{free}}$,
Gauss' law for magnetic field:	$\int \nabla \cdot \boldsymbol{B}\, d\tau = 0$,	$\nabla \cdot \boldsymbol{B} = 0$,
Faraday's law:	$\oint \boldsymbol{E} \cdot d\boldsymbol{l} = -\frac{d}{dt}\int \boldsymbol{B} \cdot d\boldsymbol{S}$,	$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$,
Ampere-Maxwell law:	$\oint \boldsymbol{H} \cdot d\boldsymbol{l} = \int \boldsymbol{j}_{\text{free}} \cdot d\boldsymbol{S} + \frac{d}{dt}\int \boldsymbol{D} \cdot d\boldsymbol{S}$.	$\nabla \times \boldsymbol{H} = \boldsymbol{j}_{\text{free}} + \frac{\partial \boldsymbol{D}}{\partial t}$.

Maxwell's equations

Maxwell's equations

+

Lorentz force formula

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}),$$

+

constitutive equations

$$\boldsymbol{B} = \mu \boldsymbol{H}, \ \boldsymbol{D} = \varepsilon \boldsymbol{E}, \ \mu = \mu_0 \mu_{\mathrm{r}}, \ \varepsilon = \varepsilon_0 \varepsilon_{\mathrm{r}},$$

constitute the basic framework of the theory of the electromagnetic field.

The first and second equations were derived for the static fields. However, they are applicable to the time-dependent situation.

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\nabla \cdot (\nabla \times \boldsymbol{E}) = -\nabla \cdot \frac{\partial \boldsymbol{B}}{\partial t}$$

$$\Downarrow$$

$$\frac{\partial}{\partial t} \nabla \cdot \boldsymbol{B} = 0 \quad \Longrightarrow \quad \nabla \cdot \boldsymbol{B} = \text{Const}.$$

$\nabla \cdot \boldsymbol{B} = 0$ for static situation

$\Rightarrow \nabla \cdot \boldsymbol{B} = 0$ for the time-dependent field.

$$\nabla \times \boldsymbol{H} = \boldsymbol{j}_{\text{free}} + \frac{\partial \boldsymbol{D}}{\partial t}.$$

$$\nabla \cdot (\nabla \times \boldsymbol{H}) = \nabla \cdot \boldsymbol{j}_{\text{free}} + \nabla \cdot \frac{\partial \boldsymbol{D}}{\partial t}.$$

$$\frac{\partial \rho_{\text{free}}}{\partial t} + \nabla \cdot \boldsymbol{j}_{\text{free}} = 0$$

$$\frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{D} - \rho_{\text{free}}) = 0. \Longrightarrow \nabla \cdot \boldsymbol{D} - \rho_{\text{free}} = \text{Const}.$$

$\nabla \cdot \boldsymbol{D} = \rho_{\text{free}}$ for static situation

$\Rightarrow \nabla \cdot \boldsymbol{D} = \rho_{\text{free}}$ for the time-dependent field.

Simple

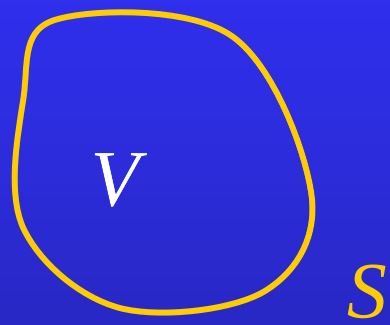
Symmetric ⟹ Beautiful

Self-consistent———Harmonic

Maxwell’s equations are compatible with the principle of the relativity- they remain invariant under a Lorentz transformation.

20.4 Energy current density and momentum density of the electromagnetic field

$$u = \frac{1}{2}\boldsymbol{E}\cdot\boldsymbol{D} + \frac{1}{2}\boldsymbol{B}\cdot\boldsymbol{H}$$



$$-\frac{dU}{dt} = -\frac{1}{2}\frac{d}{dt}\int\left(\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{B}\cdot\boldsymbol{H}\right)d\tau$$

$$= -\int\left(\boldsymbol{E}\cdot\frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{H}\cdot\frac{\partial \boldsymbol{B}}{\partial t}\right)d\tau$$

$$-\frac{\partial \boldsymbol{D}}{\partial t} = \boldsymbol{j} - \nabla\times\boldsymbol{H}$$

$$-\frac{\partial \boldsymbol{B}}{\partial t} = \nabla\times\boldsymbol{E}$$

$$-\frac{dU}{dt}=\int\left[\boldsymbol{j}\cdot\boldsymbol{E}+(\nabla\times\boldsymbol{E})\cdot\boldsymbol{H}-\boldsymbol{E}\cdot(\nabla\times\boldsymbol{H})\right]d\tau$$

$$=\int\left[\boldsymbol{j}\cdot\boldsymbol{E}+\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H})\right]d\tau$$

$$=\int\boldsymbol{j}\cdot\boldsymbol{E}\,d\tau+\oint_S\boxed{(\boldsymbol{E}\times\boldsymbol{H})}\cdot d\boldsymbol{S}.$$

Joule dissipation

Poynting vector

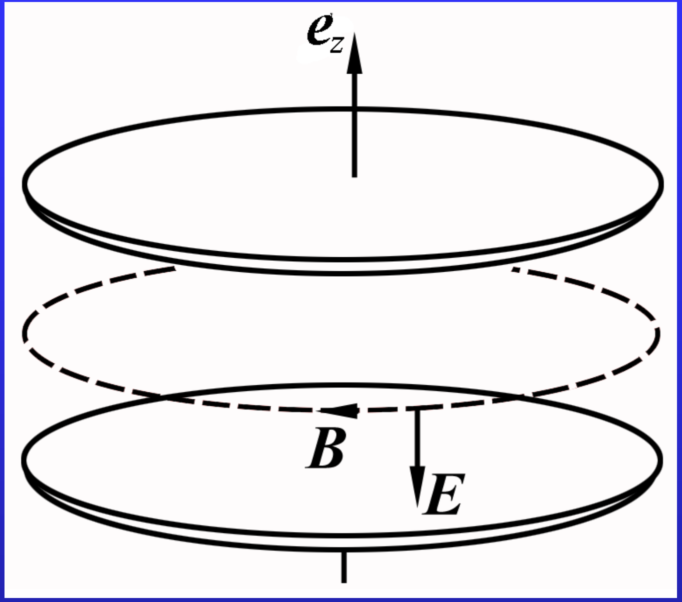
$$\boldsymbol{S}=\boldsymbol{E}\times\boldsymbol{H}$$

Energy current density vector

$$\dim S=\left(\dim\frac{dU}{dt}\right)\cdot \mathrm{L}^{-2}=\boxed{\frac{\dim U}{\mathrm{L}^3}}\dim v$$

In the vacuum, the Poynting vector is $\boldsymbol{S}=\frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B}.$

Example 20.4 Find the energy current density at the edge of a circular parallel-plate capacitor.



Solution: The electric and magnetic fields at edge of the capacitor are

$$\boldsymbol{E} = -E\boldsymbol{e}_z,$$

$$\boldsymbol{B} = \frac{1}{2}\mu_0\varepsilon_0 R\frac{dE}{dt}(-\boldsymbol{e}_\varphi).$$

Then, the Poynting vector is

$$\boldsymbol{S} = \frac{1}{2}\varepsilon_0 RE\frac{dE}{dt}(-\boldsymbol{e}_\rho).$$

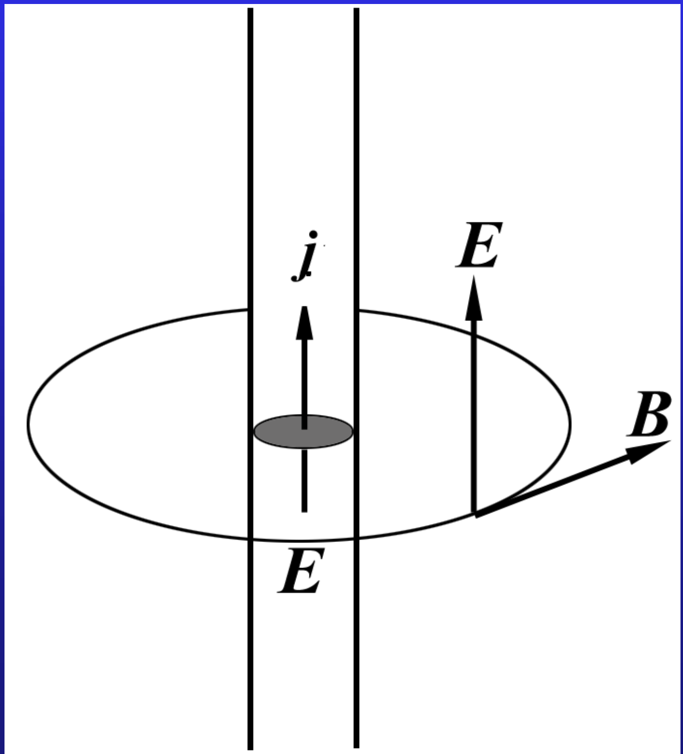
For the charging process, $dE/dt>0$, the energy flows into the capacitor. For the charging process, $dE/dt<0$, the energy flows out of the capacitor. The energy flowing into the capacitor is

$$2\pi Rd\cdot S=\pi R^2 d\varepsilon_0 E\frac{dE}{dt}=\frac{d}{dt}\left(\pi R^2 d\frac{1}{2}\varepsilon_0 E^2\right).$$

Example 20.5 Find the energy current density at the surface of a current carrying wire.

Solution: Suppose the wire is of radius r and the conductivity σ . The resistance is

$$R = \frac{1}{\sigma}\frac{l}{\pi r^2}.$$



The electric field and the magnetic field at the surface are

$$\boldsymbol{E} = \frac{\boldsymbol{j}}{\sigma} = \frac{I\boldsymbol{e}_z}{\sigma \pi r^2} = \frac{IR}{l}\boldsymbol{e}_z,$$

$$\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{2I}{r}\boldsymbol{e}_\varphi.$$

Thus, the Poynting vector is

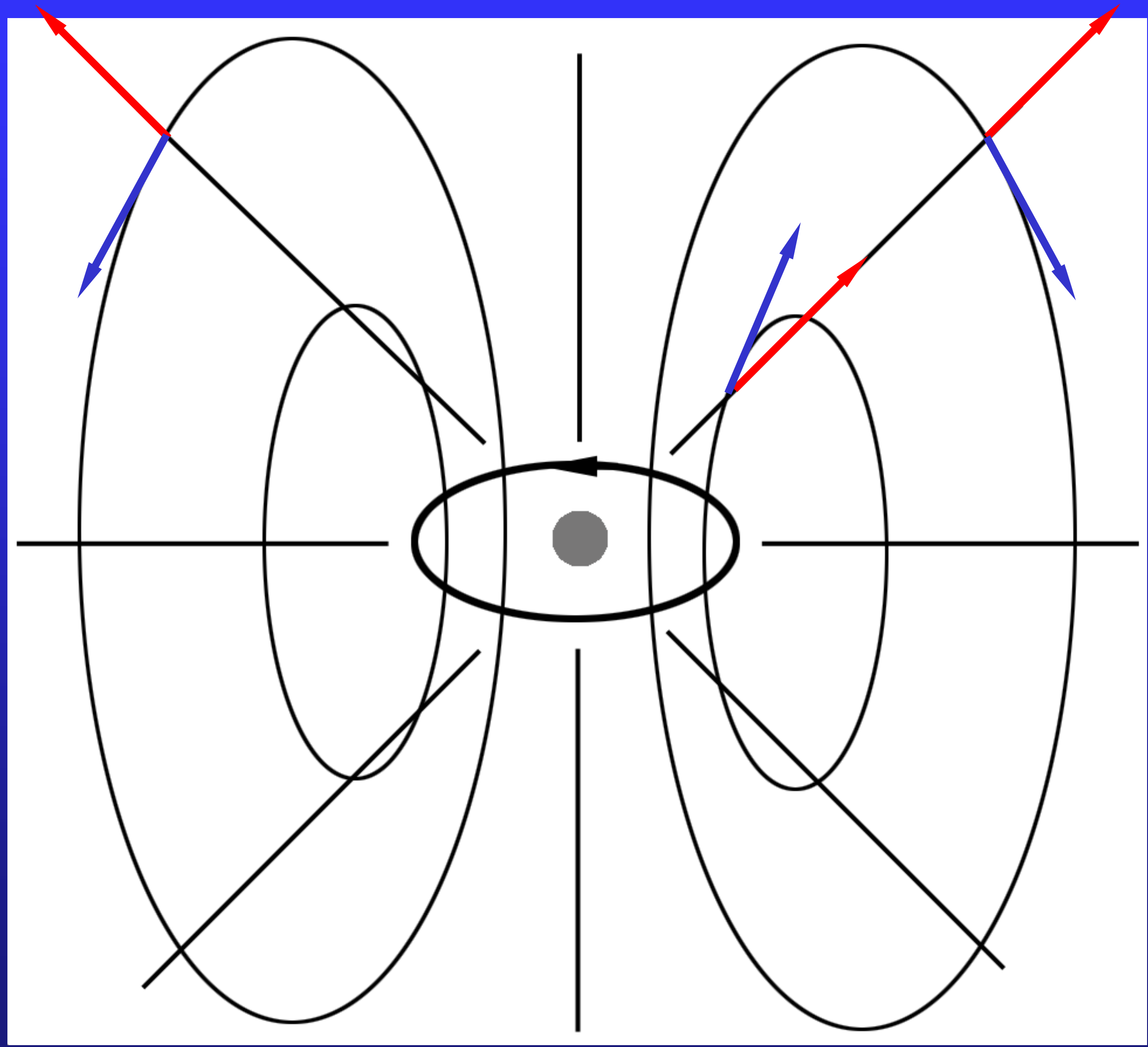
$$\boldsymbol{S} = \frac{1}{\mu_0}\frac{IR}{l}\frac{\mu_0}{4\pi}\frac{2I}{r}(\boldsymbol{e}_z \times \boldsymbol{e}_\varphi)$$
$$= \frac{I^2 R}{2\pi rl}(-\boldsymbol{e}_\rho).$$

The energy passing through the surface of a piece of wire of length l is

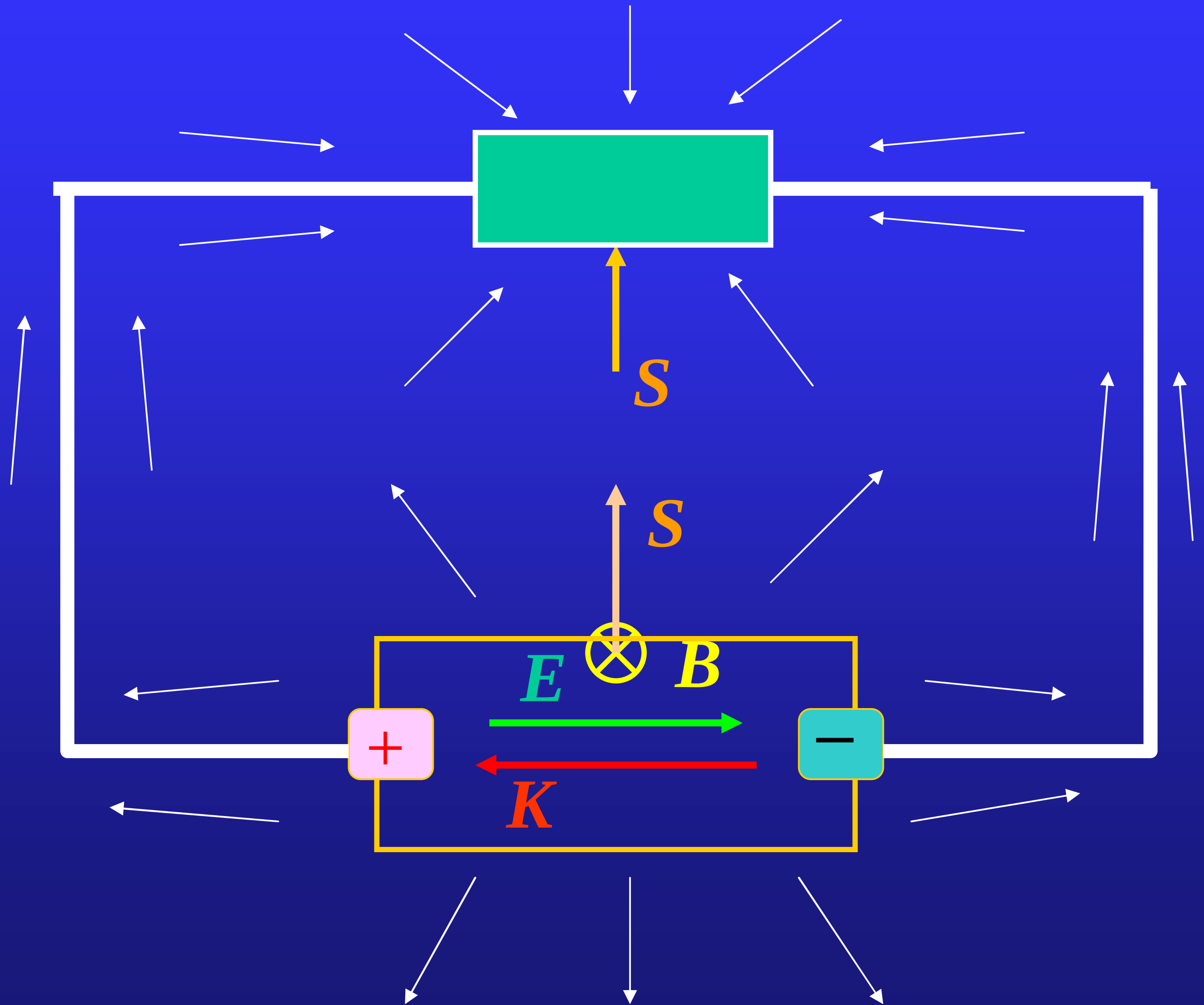
$$S \cdot 2\pi rl = -\frac{I^2 R}{2\pi rl} \cdot 2\pi rl = -I^2 R.$$

This indicates that the energy of joule heat is transferred from the outside off the wire.

Example 20.6 a point charge and a magnetic dipole



Example: Energy transportation of from the electric power supply to the electric load.



Momentum density and angular momentum density

The momentum density of electromagnetic field is

$$\boldsymbol{g} = \frac{1}{c^2}\boldsymbol{S} = \varepsilon_0(\boldsymbol{E}\times\boldsymbol{B}).$$

The angular momentum density of electromagnetic field is

$$\boldsymbol{L} = \boldsymbol{r}\times\boldsymbol{g} = \varepsilon_0\boldsymbol{r}\times(\boldsymbol{E}\times\boldsymbol{B}).$$

Assignment: 20.3(20.4E), 20.6